UAT Test Plan
for
KUNCHEE
Secure File Sharing
with E2EE

1. Objectives and Scope

1.1 Objectives and business requirements

Primary goal of this User Acceptance Test (UAT) is to verify that lecturers and staff at The Ship Campus Exam Unit can successfully and securely use the Kunchee platform to upload, manage, and share highly confidential examination materials.

We are introducing this platform to enforce strict data confidentiality, integrity, and controlled access through client-side end-to-end encryption.

We will measure success by evaluating two main factors:

1. **Security Reliability:** 100% of the uploaded files (notes, code snippets, and exam papers) must be successfully encrypted on the user's device before reaching the Firebase database and can only be decrypted by the intended authorized recipient.

2. **User Workflow:** The target users must be able to complete their core tasks (uploading an exam paper, assigning access, and downloading/decrypting the file) without experiencing workflow errors or requiring extensive technical assistance.

1.2 Scope

Currently, the Examination Unit faces significant security pain points regarding the transfer of sensitive exam materials, particularly the risks associated with physical USB transfers or unencrypted email communications.

For this UAT, we will be testing:

- The user authentication and role-based access control system (verifying the distinct permissions between Lecturers and Exam Unit Admins).
- The end-to-end encryption and decryption functionality for file uploads and downloads.
- The system's audit logging feature, ensuring user activities are tracked accurately.
- The overall user interface navigation and responsiveness on standard web browsers.

We will **not** be testing:

- Network-level security infrastructure (like campus firewalls or external Wi-Fi security).
- Password recovery via external email integrations, as credentials will be manually distributed for this testing phase.

2. Testing team

To ensure a comprehensive evaluation of the system, the testing responsibilities are divided between the development team and the actual end-users:

2.1 Testing team

Role	Assigned Personnel	Responsibilities
UAT Managers / Test Leads	Wayden, Danial, Shahridan, and Farah (Development Team)	Setting up the staging environment, distributing test accounts, guiding users, observing the testing process, and recording any technical bugs.
UAT Tester/ End User	Ms Farahzatul Shida (Head Department-Exam Unit)	Interacting with the system exactly as they would in a real-world scenario to validate that the application securely meets their daily operational needs.

Table 1 Showing testing team

3. Milestones and Deliverables

PLANNED	ACTUAL
Functional Web Application: Fully operational examination paper transfer system featuring user authentication, role-based access control, end-to-end encryption, workflow management for the examination approval chain, and audit logging capabilities.	A web-based examination paper transfer system was successfully developed. The system includes user authentication, role-based access control for different users (such as lecturers and administrators), encrypted file transfer for examination papers, and a workflow mechanism for approval within the examination chain. Audit logging was also implemented to track system activities and user actions.
System Documentation: Technical documentation including system architecture diagrams, database schema specifications, encryption workflow descriptions, API documentation, and deployment guidelines.	Comprehensive technical documentation was prepared, including system architecture diagrams, database schema design, encryption workflow explanation, API documentation, and deployment guidelines for installing and running the system.
System Analysis: Planned to implement encryption to ensure the confidentiality of examination papers during transfer and storage. End-to-end encryption was intended to protect sensitive files from unauthorized access or interception during transmission.	Encryption mechanisms were implemented to secure the transfer of examination papers within the system. Secure communication protocols were used to protect data during transmission, ensuring that files remain confidential and protected from interception. The encryption process successfully safeguarded examination documents, ensuring that only authorized users within the approval workflow could access the files.
Project Report: Final report encompassing literature review, detailed methodology, system design documentation, implementation details, testing results and project conclusions.	A complete project report was produced, containing the literature review, methodology used for system development, system design, implementation details, testing procedures, results analysis, and the conclusion of the project.

Table 2 showing deliverables

3.1 Design & testing process

Design References: Low Fidelity Sketch

Please review the provided low-fidelity sketches and attached design notes for a visual reminder of the system's layout and the 3-tier approval workflow prior to testing.

Testing Stages: The UAT process will occur in 4 distinct stages

1. **Staging Environment:** Set up by Shahridan Noriman Bin Baharud Din, this environment will closely mirror the production web application. It will include test credentials and isolated departmental data to accurately simulate the Lecturer, HOS, and Exam Unit roles.
2. **Training:** Shahridan Noriman Bin Baharud Din will conduct a comprehensive, single-session UAT training for our tester, Ms. Farahzatul Shida, on **February 12, 2026**.
3. **UAT Execution:** During a live, single-day session on **March 12, 2026**, Ms. Farahzatul Shida will execute the assigned test cases while observed by the team. She will simulate all user roles to verify end-to-end functionality, encryption, and version control. All bugs, errors, and feedback will be reported face-to-face (F2F) and documented immediately by the team on-site.
4. **Reporting:** Immediately following the live execution, the team will compile the F2F feedback for data analysis and bug triage. A final meeting will be held to review the UAT Summary Report and determine if the system is ready for university-wide deployment.

3.2 Staging environment

Access & Setup

- Staging URL: https://file-share-f8260.web.app
- Access: UAT tester will receive all test accounts with their assigned role.

Test Users: One test account per role will be given:
- Lecturers: Lecture1@test.com (Password: test123456)
- HOS: Hos1@test.com (Password: test123456)
- Exam Unit Admin: examunit@admin.com (Password: ExamUnit2025!)

Staging environment will mirror production with:

- Same Firebase project configuration
- Same encryption settings (AES-256-GCM)
- Same Cloud Functions for email notifications
- Same Firestore security rules
- Same Storage rules
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Critical Warning

Testers must NOT use staging for real exam papers. All test data must be dummy files

3.3 Training

We will be holding a comprehensive UAT training session on **February 12, 2026**.

Shahridan Noriman Bin Baharud Din will lead this session and walk Ms Farahzatul Shida through the new secure exam paper management system, explaining its core workflows and how to properly test them. To ensure a thorough understanding of the platform before university deployment, the training is broken down into the following agenda:

Session 1: Feature Overview & Business Objectives (30 minutes)

An introduction to the business goals of ensuring secure exam management with strict accountability. We will cover the 3-tier approval system workflow and key features like encryption, version control, audit trails, and notifications. This session establishes why UAT matters: ensuring the workflow is intuitive and secure.

Deliverable: Web app walkthrough.

Session 2: Staging Onboarding & Feature Deep Dive (1 hour)

Guidance on logging in with assigned test credentials and navigating role-specific dashboards. This includes a walkthrough of specific tasks: Lecturers (upload, version control, submission flow), HOS (review workflow, approval/rejection, feedback), and the Exam Unit (final approval, system admin panel). We will also demo encryption in action (file upload → encryption → Cloud Storage) and review download history/audit trails.

Deliverable: Web app navigation + quick reference guides.

Session 3: Role-Specific Best Practices (1.5 hours total)

- **Lecturer Best Practices (30 mins):** File upload standards (size, format, naming), version control workflow, responding to revision requests, and tracking approval status.
- **HOS Best Practices (30 mins):** Reviewing files in detail, providing constructive feedback, understanding approvals vs. revision requests, and utilizing department isolation (seeing only relevant department files).
- **Exam Unit Best Practices (30 mins):** Managing system-wide visibility and permissions, final approval checklists, user/department management, and reviewing audit trails for compliance.

Session 4: Bug Reporting & Feedback

- **Collection (30 minutes)** Clear instructions on how to formally report bugs, provide qualitative feedback, and correctly format test case documentation (e.g., how to accurately describe the steps taken during testing).

3.4 UAT Execution

UAT execution will be conducted as a single-day, in-person intensive session on **March 12, 2026**. During this live session, Ms. Farahzatul Shida (Head of Department-Exam Unit) will act as the sole tester, navigating the end-to-end 3-tier approval workflow (Lecturer → HOS → Exam Unit) to ensure the system flows flawlessly.

Steps:

1. **Live Onboarding & Setup:** The team will sit down with Ms. Farahzatul Shida, provide her with the test credentials for all three user roles, and ensure she is successfully logged into the staging environment.
2. **Live Test Case Execution:** She will execute her assigned test cases on the spot, switching between the Lecturer, HOS, and Exam Unit roles to simulate the full lifecycle of an exam paper.
3. **Real-Time F2F Bug Reporting:** Because this is a live session, Ms. Farahzatul Shida will report any bugs, errors, or workflow confusion directly to the team face-to-face (F2F) as they happen. A designated team member will actively document all verbal feedback and visual errors during the session.

Deadline for UAT execution: March 12, 2026 (End of Day).

UAT Success Criteria:

- All assigned test cases were executed during the live session.
- Workflow tested end-to-end (Lecturer → HOS → Exam Unit → Approved).
- Encryption verified (files uploaded are encrypted and not visible in plain text).
- Role-based access working (testers can only see respective data when logged into a specific role).
- All blocking bugs and F2F feedback successfully documented by the development team.

Reporting & Data Analysis

Because testing and feedback are occurring concurrently in person, the reporting phase will begin immediately after the live execution on March 12, 2026.

The process will involve:

- **F2F Debriefing:** A concluding face-to-face conversation with Ms. Farahzatul Shida to gather her final verdict on usability, workflow clarity, and her confidence in the system's security.
- **Bug Triage & Logging:** The team will compile all the manual notes, verbal feedback, and bugs witnessed during the live session into a master **UAT Summary Report**.
- **Action Plan:** The report will categorize the F2F feedback into critical bug fixes (must be fixed before launch) and UI/UX improvements, forming the final development checklist before university deployment.

3.5 Reporting & Data analysis

1. Analysis of Test Cases & Struggles

During the live execution on March 12, 2026, Ms. Farahzatul Shida executed all assigned test cases across the Lecturer, HOS, and Exam Unit roles. The core 3-tier approval workflow and AES-256-GCM encryption functioned as expected. However, the tester struggled primarily with the revision request loop. Specifically, when a paper was rejected by the HOS, the UI did not clearly indicate to the simulated Lecturer role where to upload the newly revised document, causing a minor workflow bottleneck.

2. General Feedback

The overall feedback from the Head of Department was highly positive, particularly regarding the system's security features. The isolated departmental views and the automated audit trails provided a strong sense of accountability. The tester noted that the system successfully replaces the manual, insecure physical handling of exam papers with a streamlined digital process.

3. Areas of Improvement

Based on the live session, the immediate areas of improvement required before university-wide deployment include:

- **UI/UX Fix:** Redesigning the "Revision Required" dashboard to make the re-upload button more prominent for lecturers.
- **Bug Fix:** Resolving a critical bug where the final submit button occasionally disables for the Exam Unit role.
- **Performance:** Optimizing the loading time for large PDF files during the initial upload phase.

Bug Tracking

Bug ID	Title	User Role	Severity	Status	Fix Assigned	Notes / Live Observation
UAT-001	Approval status does not update immediately after clicking	HOS	Medium	Resolved	Shahridan	The tester thought the system froze. The page had to be refreshed to see the 'Approved' status. This React state management issue was fixed on the spot.
UAT-002	Laptop freezes briefly when decrypting large exam papers	Exam Unit Admin	Low	Open	Dev Team	The tester tried to decrypt a 15MB+ PDF. The browser lagged for about 4 seconds because client-side encryption processing is heavy on standard RAM. This is an acceptable system limitation.
UAT-003	Unclear status of the document in the approval timeline	Lecturer	Minor	Open	Danial	After uploading, the tester was unsure whose turn it was to review the file (HOS or Exam Unit). We will add a simple visual timeline progress bar (e.g., 'Pending HOS Review') in the next UI update.

Table 3 Showing bug tracking

Feedback Summary Report

Category	Qualitative Feedback (Verbal)	Priority
Security	*"This system is great. The exam questions turn into scrambled text (encrypted) before even entering the database. I feel much less worried about leaks compared to using physical USB drives."*	High
Usability	*"The approval flow from Lecturer to HOS perfectly matches our current workflow. However, I got a bit confused trying to find the 'Upload Revised Version' button. Could you make that button bigger?"*	Medium
Performance	*"It is fast for normal files. But earlier, when testing a heavy PDF, it took a while to load during decryption. Could you add a 'Loading spinner' so we know the system is still processing and hasn't crashed?"*	Medium
Feature Request	*"Everything works well, but in the future, I would like the system to send an automated email when a new file needs my approval. I cannot be expected to check the website every single day."*	Low

Table 4 Showing feedback summary

Analysis Meeting Detail

Item	Details
Date	12 March 2026
Time	2:00 PM
Attendees	Development Team & Exam Unit

GO to Production if:

Criteria	Status
No critical bugs remaining	Pass
All high severity bugs have workarounds or scheduled fixes	Pass
End-to-end workflow tested successfully	Pass
Security and encryption validated by tester	Pass

NO-GO (Delay Release) if:

Issue	Impact
Critical bugs blocking workflow	System unusable
Security vulnerabilities (e.g., exposed encrypted files)	High risk
Multiple users unable to complete tasks	Deployment failure

4. Environmental Requirements

To ensure the User Acceptance Testing (UAT) is conducted smoothly and reflects real-world usage at The Ship Campus Exam Unit, the following hardware, software, and test environment setups are required.

4.1 Hardware Requirements

The UAT must be executed on standard office equipment that matches what the staff and lecturers currently use on campus.

Component	Minimum Requirement	Justification
Device Type	Standard Desktop PC or Laptop	Represents the typical daily workstation used by the academic staff and Exam Unit members.
Processor	Intel Core i3 / AMD Ryzen 3 (or equivalent)	Ensures the device has enough processing power to handle the complex mathematical calculations required for client-side encryption without freezing.
Memory (RAM)	4 GB RAM (8 GB recommended)	Required to maintain smooth browser performance while encrypting or decrypting large examination files locally.
Network	Stable broadband or Campus Wi-Fi	A continuous internet connection is strictly required to authenticate users and sync encrypted files with the Firebase database.

Table 5 Showing hardware requirements

4.2 Software Requirements

Since Kunchee is a web-based application utilizing end-to-end encryption, the software environment relies heavily on modern browser capabilities rather than third-party installations.

Component	Requirement	Justification
Operating System	Windows 10 / Windows 11, or macOS	Ensures compatibility with the standard operating systems provided by the university's IT department.
Web Browser	Google Chrome, Mozilla Firefox, or Microsoft Edge (Latest Versions)	**Crucial:** The browser must natively support the **Web Crypto API**. Outdated browsers (like Internet Explorer) will fail to execute the client-side encryption protocols.
Additional Software	None required	The required cryptographic libraries (libsodium.js) are integrated directly into the web application. Users do not need to install software like WinRAR or external VPNs.

Table 6 Showing software requirements

4.3 Test Environment Setup

To protect actual university data, the UAT will be conducted in a secure, isolated testing environment.

Setup Item	Description
System URL	The testing will be conducted on a dedicated staging link
Database Environment	A separate Firebase Test Database will be used. This ensures that any dummy files uploaded during testing do not mix with real, confidential exam papers.
Test Accounts	The development team will provide pre-configured, dummy accounts with specific roles assigned (e.g., *Test_Lecturer_01*, *Test_ExamAdmin_01*) to verify the role-based access controls safely.
Test Data	Tester will be provided with sample PDF documents (labelled "DUMMY EXAM PAPER") and sample text files to upload, ensuring no actual confidential materials are exposed during the test.

Table 7 Showing test environment setup

5. Features to be tested

5.1. Client-Side Encrypted File Upload and Sharing (Lecturer Workflow)

5.1.1. Pass/fail criteria

- **Pass**: The platform successfully encrypts the sample file locally on the device (client-side) before upload, securely stores it in the database, and records the action in the audit log.
- **Fail**: The system uploads the file unencrypted, the browser freezes during the encryption calculation, or the system fails to log the upload in the audit tracker.

5.1.2. Test cases

1. Log on to the provided Kunchee staging URL using the assigned test account: Test_Lecturer_01.
2. Navigate to the file upload module on the dashboard.
3. Select and upload the sample PDF document labeled "DUMMY EXAM PAPER".
4. Assign specific access permissions for the uploaded file exclusively to Test_ExamAdmin_01.
5. Submit the file and verify that a success notification appears confirming that the encrypted upload is complete.

5.2. Role-Based Access Control and Secure Decryption (Admin Workflow)

5.2.1. Pass/fail criteria

- **Pass**: The Exam Unit Admin can successfully view the shared exam paper on their dashboard, download it, and decrypt it locally to read the original contents without errors.
- **Fail**: The system denies the authorized admin access, the local decryption process fails, or the downloaded file is corrupted or unreadable.

5.2.2. Test cases

1. Log on to the provided Kunchee staging URL using the assigned test account: Test_ExamAdmin_01.
2. Navigate to the shared files or inbox module.
3. Locate the "DUMMY EXAM PAPER" previously shared by Test_Lecturer_01.
4. Click the download button to initiate the client-side decryption process.
5. Open the downloaded PDF on the local machine and verify that the contents are perfectly intact and legible.

5.3. Features to avoid testing

5.3.1. Network-Level Security Infrastructure

Tester must not attempt to test, bypass, or stress-test campus firewalls or external Wi-Fi security. The scope of this UAT is strictly evaluating Kunchee's application-layer encryption and lecturer workflow, not the university's network perimeter defense system.

5.3.2. Password Recovery and Email Integrations

Tester must not test the "Forgot Password" or external email notification workflows. Because all test accounts for this phase have been pre-configured and manually distributed by the development team, automated email recovery systems are out of scope for this session.